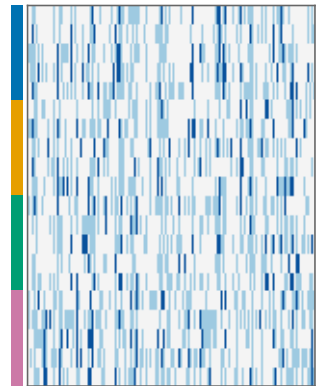
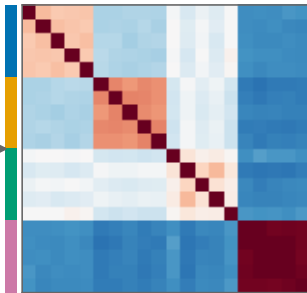


genotypes G



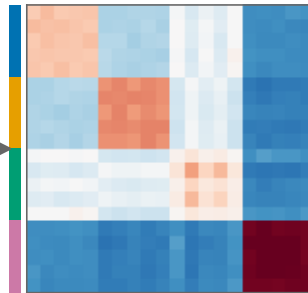
centre +
standardise

standard GRM
(centred, scaled)



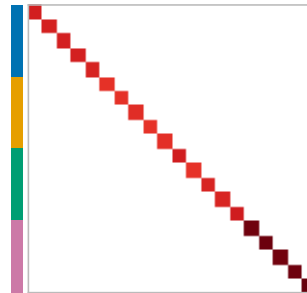
PCP
 $C = L + S$

L : diag. imputed,
 $\min \sum_k |\lambda_k|$ (all eigenvectors)



+

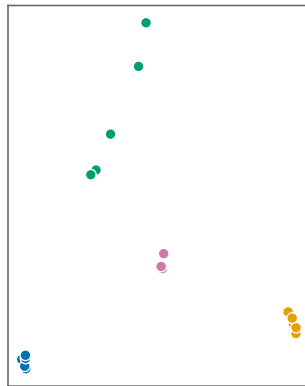
S : the diagonal



eigen-
vectors

PC3

PCs: PC2 vs PC3
MXL axis $R^2 = 0.99$



PC2